In the name of God

HomeWork #1 for “Digital Image processing (DIP)” course

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Farvardin 18, 1405, Deadline: Ordibehesh 1, 1405, Midnight time

Each question has 5 points

1- Define a digital image mathematically. If an image has spatial resolution $512 \times 512$and intensity resolution of 256 levels, how many bits are required to store the image?

2- An imaging system captures intensity values between 0 and 4095. Determine the number of bits required and the dynamic range. Explanation: Dynamic range in digital image processing is the ratio between the maximum and minimum measurable intensity values in an image or imaging system. High dynamic range means more intensity levels, better detail in both dark and bright regions. Low dynamic range means loss of detail due to saturation or poor contrast. Mathematically: Dynamic Range $= \frac{I_{max}}{I_{min}}$

3- A color image has resolution 1920×1080 with 24 bits per pixel. Compute the storage size.

4- An image contains spatial frequency components up to 20 cycles/mm. According to Nyquist sampling theorem, determine the minimum sampling frequency. Explanation: The Nyquist Sampling Theorem states that:
A continuous signal can be perfectly reconstructed from its samples if it is sampled at a rate at least twice the highest frequency present in the signal. If the smallest visible object detail corresponds to 0.5 mm, the sampling interval should be: ≤ 0.25 mm to satisfy the Nyquist condition.

5- If an image intensity range is 0–255 and it is quantized to 16 levels, calculate the quantization step size.

6- Two images have resolutions $512 \times 512$and 1024×1024. How many times more pixels do the second image contain?

7- An image originally uses 10-bit intensity resolution. If reduced to 8-bit, how many intensity levels are lost?

8- An image sensor has size $10\ mm \times 10\ mm$ with resolution 2000×2000. Calculate pixel density (pixels/mm).

9- If a scanner samples an image at 300 dpi, how many pixels are used to represent a 5 inch × 7-inch photograph?

10- If the sampling frequency is 500 samples/mm, determine the sampling interval.

11- Explain the two steps required to convert a continuous image into a digital image.

12- A grayscale image of size 4096×4096 uses 12 bits per pixel. Compute required storage.

13- What happens if sampling frequency is lower than Nyquist rate?

14- An image has 64 gray levels. Determine the number of bits required.

15- If an image uses 7-bit quantization, determine the number of gray levels.

16- A video frame has resolution 1280×720 with 8-bit grayscale. Calculate memory per frame.

17- If an image is sampled at 600 dpi and reduced to 300 dpi, how does the number of pixels change?

18- If quantization step size is $\Delta = 10$, compute maximum quantization error.

19- An image has dimensions 1920×1080. Determine the aspect ratio.

20- A digital imaging system captures images of size 10 cm × 10 cm. The smallest visible detail in the scene is 0.25 mm.

a) Determine the minimum sampling rate (pixels per cm) required to satisfy the Nyquist sampling theorem.
b) Compute the minimum image resolution (number of pixels) required.
c) If the image is quantized using 8 bits per pixel, calculate the storage required for one image.

All answers for each HomeWork should be given:

- In one file, (at the top of file: Student name, student number, HW#1, DIP)
- Format: PDF (typed)
- Name of file: student's name_student number (For example: David_Peter_2020234.pdf)
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